Semestertoets 2	Semester Test 2
Kopiereg voorbehou	Copyright reserved
Elektrisiteit en Elektronika EBN 122	**Electricity and Electronics EBN 122**
1 Oktober 2018	1 October 2018

Toetsinligting - Test Information			
Maksimum punte: Maximum marks:	45	Volpunte: Full marks:	46
Duur van vraestel: Duration of paper:	90 min + (10 for OCR)	Oop-/Toeboek: Open/Closed book:	Toe/Closed
Totale aantal bladsye (hierdie blad ingesluit): Total number of pages (this page included)		16	

Important

1. The examination regulations of the University of Pretoria apply.
2. All questions must be answered on the Optical Character Recognition (OCR) form.
3. Read the instructions on the OCR form carefully.
4. Where applicable, answers must include units.
5. Final answers must be written as real numbers and not as fractions.

Belangrik

1. Die eksamenregulasies van die Universiteit van Pretoria geld.
2. Alle vrae moet op die Optiese Karakter Herkenning (OKH) vorm ingevul word.
3. Neem noukeurig kennis van die instruksies op die OKH vorm.
4. Indien van toepassing, moet antwoorde eenhede insluit.
5. Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.

Lecturers/Dosente:	D. le Roux, J.P. de Villiers, J.P Jacobs

INSTRUCTIONS

Do step 0 in the first 5 minutes of the session. (-5 to 0 min)

Do step 1 in the first 90 minutes of the test. (0-90 min)

Do step 2 within the following 10 minutes. (90-100 min)

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form.

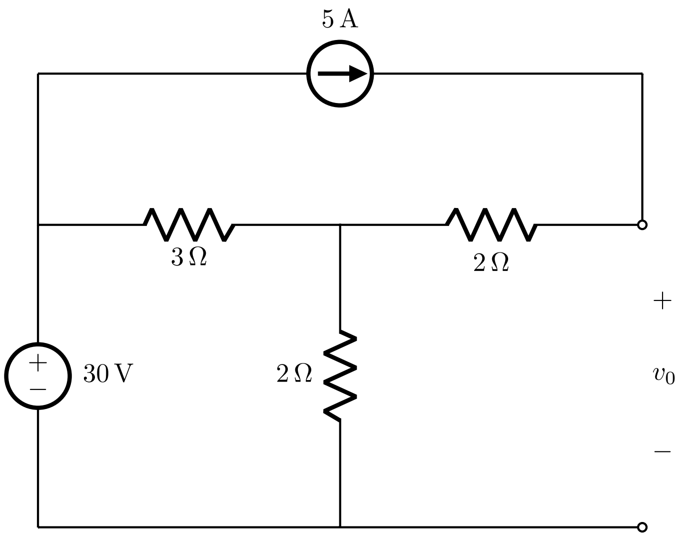
STEP 1 (0 min to 90 min): Do your own calculations on the QUESTION PAPER. The question paper will not be taken in at the end of the test. Write down the answers for each question on the PRACTICE ANSWER SHEET. Be sure to include the units. The PRACTICE ANSWER SHEET will not be taken in at the end of the test.

STEP 2 (90 min to 100 min): Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

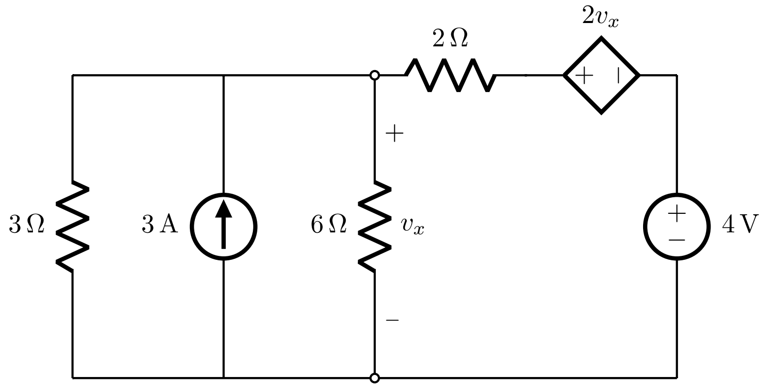
- The Assessment ID is: **2018EBN122S2**
- Use a mechanical pencil with 0.5mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops as decimal points.

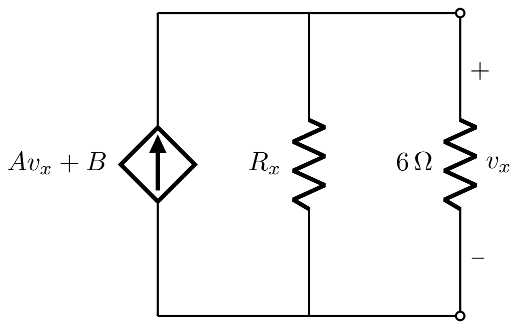
Section 1 / Afdeling 1

Questions 1 to 2 [4]		Vrae 1 tot 2 [4]	
Find the contribution to the voltage v_0 due to:		Bepaal die bydrae tot die spanning v_0 as gevolg van:	
1	the 30 V source [2]	**1**	die 30 V bron [2]
2	the 5 A source [2]	**2**	die 5 A bron [2]

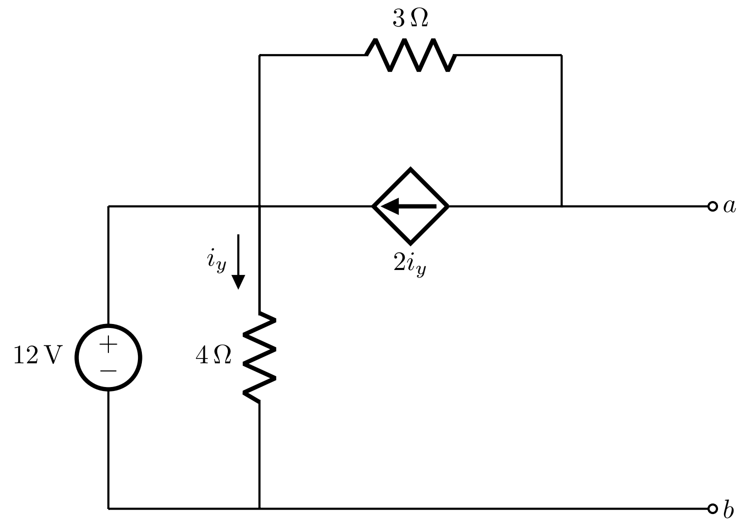


Questions 3 to 5 [4]		Vrae 3 tot 5 [4]	
Simplify the top circuit by means of source transformation so that it looks like the bottom circuit. Determine:		Vereenvoudig die boonste stroombaan met behulp van brontransformasie sodat dit lyk soos die onderste stroombaan. Bepaal:	
3	R_x [1]	**3**	R_x [1]
4	A [1]	**4**	A [1]
5	B [2]	**5**	B [2]

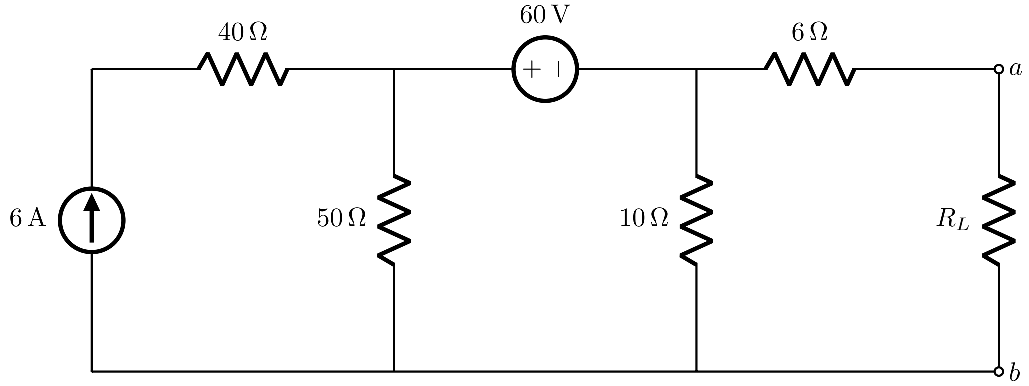




Questions 6 to 7 [4]		**Vrae 6 tot 7 [4]**	
Determine the Norton equivalent current I_N and resistance R_N with reference to terminals $a-b$.		Bepaal die Norton ekwivalente stroom I_N en weerstand R_N met verwysing tot terminale $a-b$.	
6	I_N [2]	**6**	I_N [2]
7	R_N [2]	**7**	R_N [2]



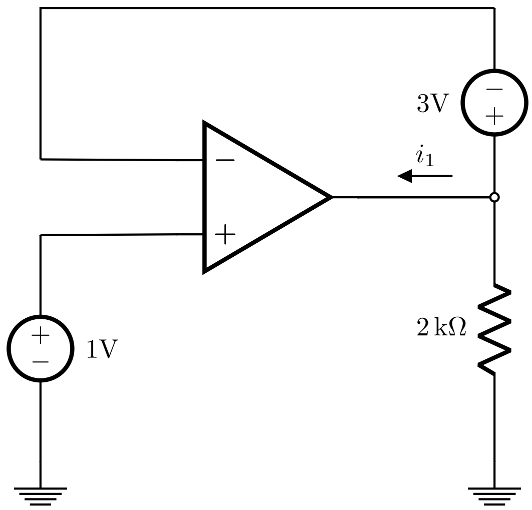
Questions 8 to 9 [4]		Vrae 8 tot 9 [4]	
Find the Thevenin equivalent voltage V_{Th} with respect to terminals a–b, and the value of the load resistor R_L so that maximum power is transfered to R_L.		Bepaal die Thevenin ekwivalente spanning V_{Th} met betrekking tot terminale $a-b$, en die waarde van die lasweerstand R_L sodat maksimum drywing oorgedra word aan R_L.	
8	V_{Th} [2]	**8**	V_{Th} [2]
9	R_L [2]	**9**	R_L [2]



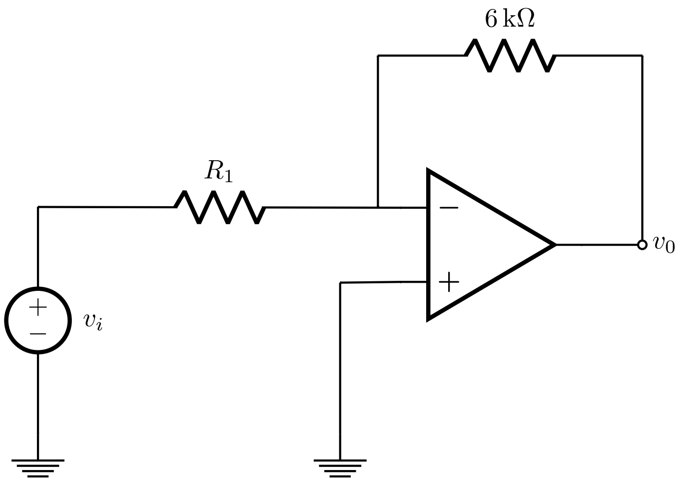
Total Section 1 / Totaal Afdeling 1: [16]

Section 2 / Afdeling 2

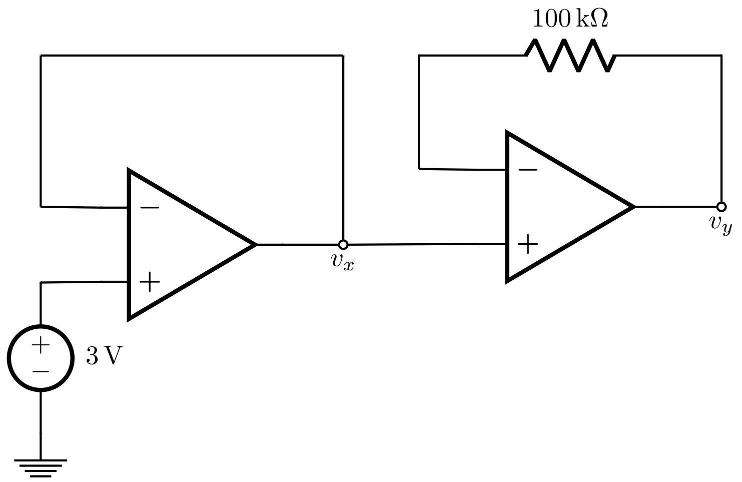
Question 10 [2]		**Vraag 10 [2]**	
Assume that the op-amp is ideal. Find i_1.		Aanvaar dat die OV ideaal is. Bepaal i_1.	
10	i_1 [2]	**10**	i_1 [2]



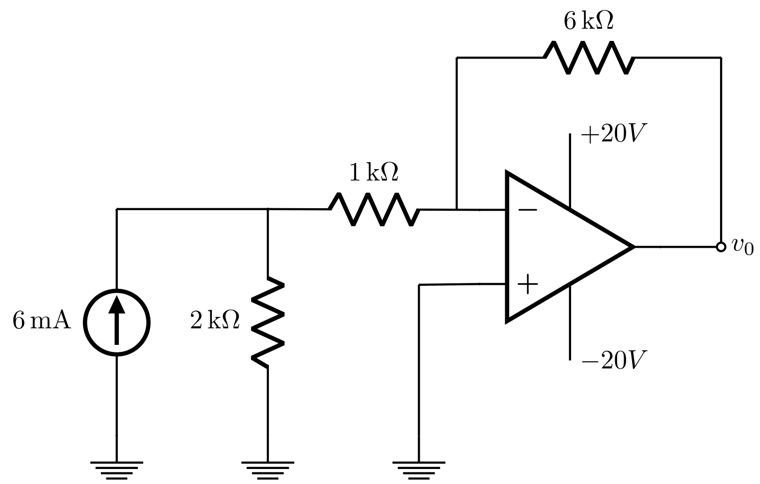
Question 11 [2]		**Vraag 11 [2]**	
Assume that the op-amp is ideal. Find R_1 so that the ratio v_0/v_i will be equal to -1.5.		Aanvaar dat die OV ideaal is. Bepaal R_1 sodanig dat die verhouding v_0/v_i gelyk sal wees aan -1.5.	
11	R_1 [2]	**11**	R_1 [2]



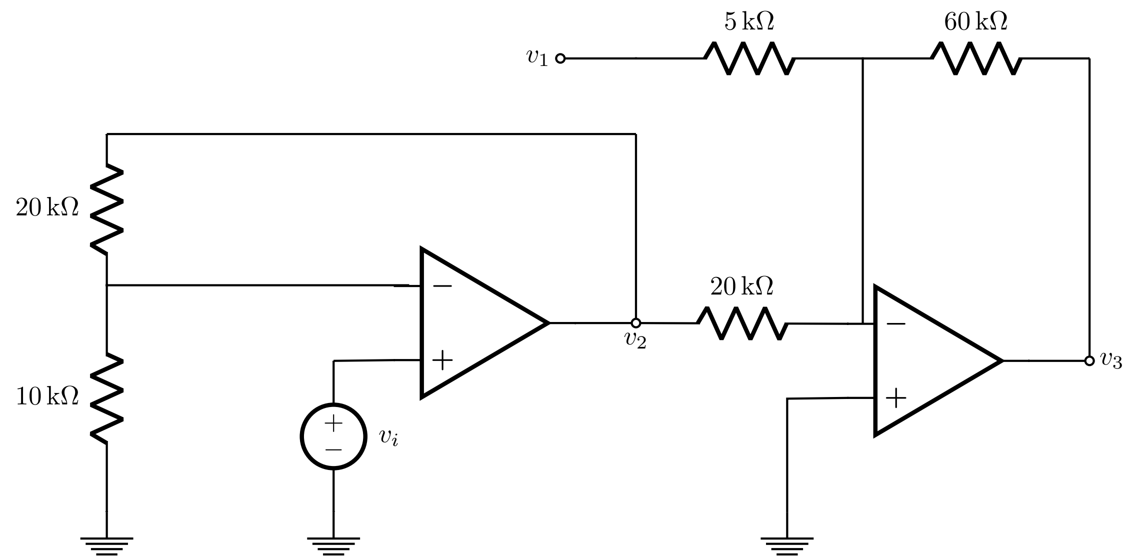
Questions 12 to 13 [2]		**Vrae 12 tot 13 [2]**	
Assume that the op-amps are ideal. Find v_x and v_y.		Aanvaar dat die OVs ideaal is. Bepaal v_x en v_y.	
12	v_x [1]	**12**	v_x [1]
13	v_y [1]	**13**	v_y [1]



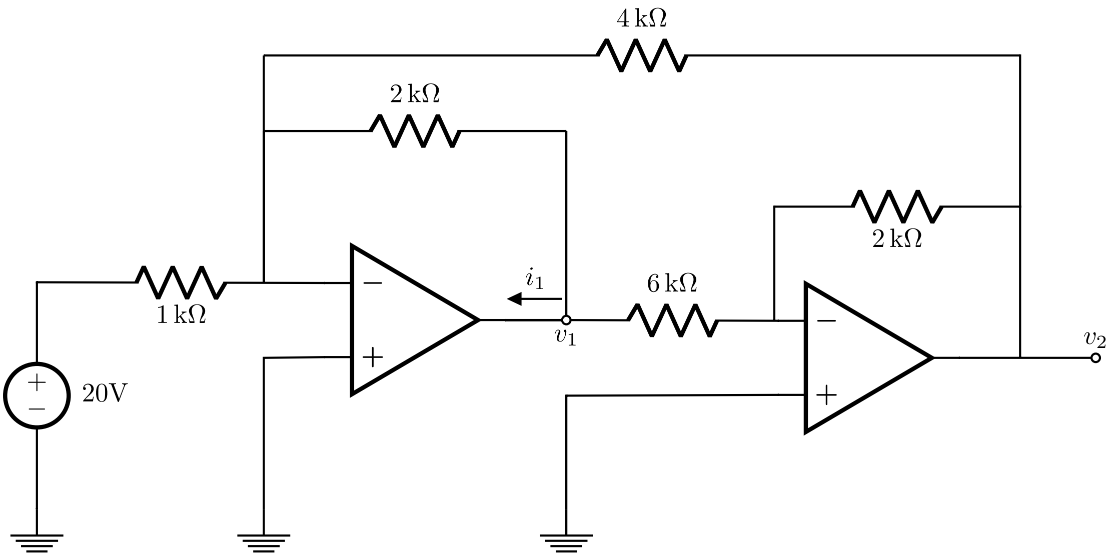
Question 14 [2]		**Vraag 14 [2]**	
Assume that the op-amp is ideal. Find v_0.		Aanvaar dat die OV ideaal is. Bepaal v_0.	
14	v_0 [2]	**14**	v_0 [2]



Questions 15 to 16 [4]		**Vrae 15 tot 16 [4]**	
Assume that the op-amps are ideal. It is known that $v_1 = -2V$ and $v_2 = 30V$. Find v_i and v_3.		Aanvaar dat die OVs ideaal is. Dit is bekend dat $v_1 = -2V$ en $v_2 = 30V$. Bepaal v_i en v_3.	
15	v_i [2]	**15**	v_i [2]
16	v_3 [2]	**16**	v_3 [2]



Questions 17 to 18 [4]		**Vrae 17 tot 18 [4]**	
Assume that the op-amps are ideal. It is known that $v_1 = -24V$. Find i_1 and v_2.		Aanvaar dat die OVs ideaal is. Dit is bekend dat $v_1 = -24V$. Bepaal i_1 en v_2.	
17	i_1 [2]	**17**	i_1 [2]
18	v_2 [2]	**18**	v_2 [2]

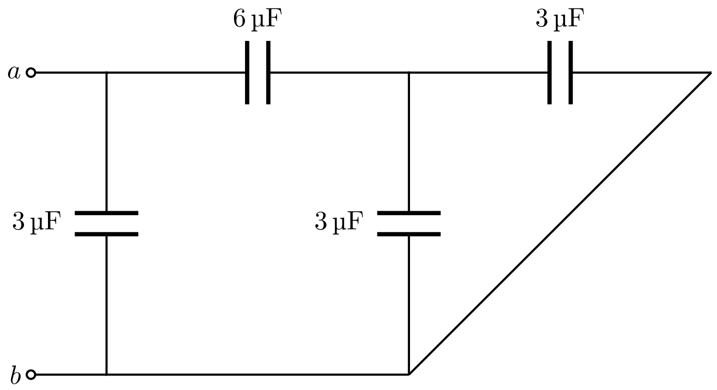


Total Section 2 / Totaal Afdeling 2: [16]

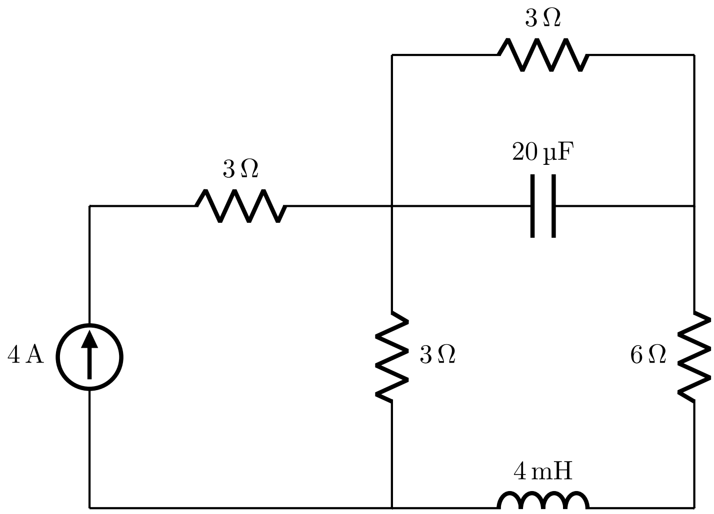
Section 3 / Afdeling 3

Question 19 [2]		Vraag 19 [2]	
Calculate the voltage at $t = 10$ ms across a 2μF capacitor if the current through it is given by $i(t) = 4e^{-2000t}$ mA. Assume that $v(0) = 500$ mV.		Bereken die spanning by $t = 10$ ms oor 'n 2μF kapasitor indien die stroom deur die kapasitor gegee word deur $i(t) = 4e^{-2000t}$ mA. Aanvaar dat $v(0) = 500$ mV.	
19	$v(10$ ms$)$ [2]	**19**	$v(10$ ms$)$ [2]

Question 20 [2]		**Vraag 20 [2]**	
Find the equivalent capacitance C_{eq} with respect to terminals $a - b$.		Bepaal die ekwivalente kapasitansie C_{eq} met verwysing tot terminale $a - b$.	
20	C_{eq} [2]	**20**	C_{eq} [2]



Questions 21 to 22 [4]		**Vrae 21 tot 22 [4]**	
Under dc conditions, calculate the energy stored in the capacitor (w_C) and the energy stored in the inductor (w_L).		Onder gelykstroomtoestande, bepaal die energie gestoor in die kapasitor (w_C) en die energie gestoor in die induktor (w_L).	
21	w_C [2]	**21**	w_C [2]
22	w_L [2]	**22**	w_L [2]



Total Section 3 / Totaal Afdeling 3: [8]

Section 4 / Afdeling 4

Questions 23 to 29 [6]		**Vrae 23 tot 29 [6]**	
Consider the signals $v_1(t) = 3\cos(20\pi t + 30°)$, and $v_2(t) = \sin(20\pi t - 60°)$. a. Determine $v = v_1 + v_2$. Write your final answer in the form $A\cos(\omega t+\phi)$, where $A > 0$ and $-180° \leq \phi \leq 180°°$. [2] b. Find the angular frequency of v_1. [1] c. Find the frequency of v_2. [1] d. Which statement is correct: 1. v_1 leads v_2 2. v_1 lags v_2 [1] e. Find the phasor $\mathbf{V_2}$ of v_2, and write your answer in rectangular form, i.e. $C + jD$. [1]		Beskou die seine $v_1(t) = 3\cos(20\pi t + 30°)$, en $v_2(t) = \sin(20\pi t - 60°)$. a. Bepaal $v = v_1 + v_2$. Skryf jou finale antwoord in die vorm $A\cos(\omega t + \phi)$, waar $A > 0$ en $-180° \leq \phi \leq 180°°$. [2] b. Bepaal die hoekfrekwensie van v_1. [1] c. Bepaal die frekwensie van v_2. [1] d. Watter stelling is korrek: 1. v_1 is voorlopend tot v_2 2. v_1 is nalopend tot v_2 [1] e. Bepaal die fasor $\mathbf{V_2}$ van v_2, en skryf jou antwoord in reghoekige vorm, i.e. $C + jD$. [2]	
23	A [1]	**23**	A [1]
24	ϕ [1]	**24**	ϕ [1]
25	angular frequency [1]	**25**	hoekfrekwensie [1]
26	frequency [1]	**26**	frekwensie [1]
27	choose 1 or 2 [1]	**27**	kies 1 of 2 [1]
28	C [0.5]	**28**	C [0.5]
29	D [0.5]	**29**	D [0.5]

Total Section 4 / Totaal Afdeling 4: [6]

Grand Total / Groottotaal : [46]

PRACTICE PAGE / OEFENBLAD

Assessment ID	2	0	1	8	E	B	N	1	2	2	S	2		

How to enter special units:										
6	0	°								
1	8	.	5	6		r	a	d	÷	s
1	5	0				H	z			
Use ÷ instead of /										

	Answer (please round all answers to 5 significant digits)	Unit
01	30 V source:	
02	5 A source:	
03	$R_x =$	
04	$A =$	No unit required
05	$B =$	No unit required
06	$I_N =$	
07	$R_N =$	
08	$V_{Th} =$	
09	$R_L =$	
10	$i_1 =$	
11	$R_1 =$	
12	$v_x =$	
13	$v_y =$	
14	$v_0 =$	
15	$v_i =$	
16	$v_3 =$	
17	$i_1 =$	
18	$v_2 =$	
19	$v(10\text{ms}) =$	
20	$C_{eq} =$	
21	$w_C =$	
22	$w_L =$	
23	$A =$	No unit required
24	$\phi =$	
25	Angular frequency:	
26	Frequency:	
27	Choose 1 or 2:	No unit required
28	$C =$	No unit required
29	$D =$	No unit required